

Colloquium Defence: Attacking the Experiment from Every Angle

A Unified Benchmark of Grokking Predictors in Neural Networks — review of Predictors 1 to 5 (L2 Norm, Dropout, Spectral, AGE, HTSR Alpha) on the 5-seed Nanda-Unified run

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Sources: `context.md` (read in full), `06_Code/`, `08_Experiments/results/nanda_unified/` at commit `4e4390a`.

No code, config or result file was changed for this report.

0. Purpose of this report and how to read it

The aim here is simple. We play the examiner. For every part of the work already done, we ask: "From which side can someone attack this?" Nothing new is invented. No new predictor, no new detection rule, no new signal is proposed as a contribution. Where a check is suggested, it is a check of the existing work, not an extension of it.

Each attack angle is written in the same five parts:

- **Attack** — the question an examiner could ask, in his words.
- **Where we stand** — what the project record and the data actually say today.
- **Risk** — High, Medium or Low, and status: Open, Partly defended, or Defended.
- **What to do** — the smallest check or wording change that closes the gap.
- **Suggested answer** — a short line you can say in the colloquium.

Three small read-only checks were run on the saved results while preparing this report (Section 9). Two of them changed my own reading of the Dropout and Spectral results. Please read those findings before the colloquium, because they are the most exposed points.

Honesty note. The quick checks in Section 9 were computed today from saved `.npy` files with a few lines of NumPy. They are not part of the pipeline, not committed, and not yet recorded in `context.md`. They are descriptive. They do not change any recorded verdict by themselves. They show where an examiner could push, and they need to be re-done properly inside the project before being quoted in the thesis.

1. One-page summary: the ten most dangerous angles

#	Attack angle	Risk	Status	Cost to close
1	No positive control: every predictor fails, so how do we know the protocol can pass anything at all?	High	Open	Low (checkpoint-only)
2	Dropout-variance peak is almost fully explained by binomial sampling of the mean dropout accuracy (Section 9.1)	High	Open	Low (saved data)
3	Spectral k_{90} is decided inside the null space of a rank- ≤ 128 kernel, so its value (≈ 3400) is not meaningful; and the representation spectrum does collapse around grok (Section 9.2)	High	Open	Low (checkpoint-only)
4	"Pre-registered criteria" claim does not match the history in <code>context.md</code> (criteria changed three times)	High	Open	Zero (wording)
5	The "extreme value" event rule is imposed by us; most source papers never define a timing rule. This clashes with the "published method only" scope	High	Partly defended	Zero (wording) + optional AUC
6	Extreme-value events depend on when training stops (horizon dependence, Section 9.3)	Medium	Open	Low (saved data)
7	Only 5 seeds; seed changes both the data split and the initialisation; seed 4 drives the correlations	Medium	Partly defended	Medium (retraining)
8	One task, one prime, one train fraction, one optimiser setting: can the verdict generalise?	Medium	Partly defended	High (retraining)
9	Canonical comparators are missing: Nanda's own progress measures, Notsawo et al. (2023), Kumar et al. (2024), Liu et al. Omnigrok	Medium	Open	Low to medium
10	Definitional choices per predictor (NC1 form, HTSR on small matrices with $\alpha < 2$, dropout placement, L2 rules being our own design)	Medium	Partly defended	Low (checkpoint-only)

The good news first. The core finding (no predictor, as implemented, fires ahead of grok and tracks it across seeds) is not in danger from most angles. Many attacks can be closed from the 801 saved checkpoints per seed, without any retraining. The dangerous part is not the result. The dangerous part is some of the *explanations* and *wordings* written around the result.

2. Angles on framing and claims

2.1 "Your protocol rejects everything. Can it accept anything?"

High

Open

Attack. Five out of five predictors fail. An examiner will say: maybe the test is simply impossible to pass. A benchmark that no signal can pass tells us about the benchmark, not about the predictors.

Where we stand. The defence note of 17 September says: "if every candidate had passed, that would be a warning sign of too-loose criteria." The mirror argument is also true. If every candidate fails, the examiner can say the criteria are too tight. At present there is no positive control on record.

What to do. Run the same two tests on one or two signals that we already know are tied to grokking, only to show that the test *can* be passed. These are not new predictors; they are sanity checks. Candidates, all computable from saved checkpoints: (a) test loss (an oracle, since it uses test labels), and (b) Nanda et al.'s own progress measures, restricted and excluded loss, if time allows. Also add a trivial negative control: "grok happens at a fixed epoch equal to the mean grok epoch". If test loss passes and the trivial rule fails, the protocol is shown to discriminate.

"The protocol is not a wall. As a positive control, test loss passes both tests on the same five seeds, so a signal that is genuinely tied to the transition does pass. The five predictors fail because their signals do not lead the transition."

Say this only after the positive-control check is actually done.

2.2 "You say the criteria were pre-registered. Were they?"

High

Open

Attack. The 17 September defence note says the falsification test "was pre-registered before Predictor 1 was run and applied unchanged through Predictor 4". An examiner who reads the repository history will find otherwise.

Where we stand. The record shows the criteria evolving:

- 7 August: three criteria (always before grok; small and consistent gap as a proportion of the run; clearly above the noise floor).
- 7 September (v4): two criteria (extremum leads grok; failure direction consistent across seeds).
- 26 September: two criteria (fires before grok in 5/5; Spearman ρ across seeds, significant only at $\rho \geq 0.9$ for $n = 5$).

The third criterion (noise floor) was never formally applied to Predictors 2 to 5. Also, L2 Norm's detection rules were tuned (five strategies) before the "no tuning" scope rule existed.

What to do. Remove the word "pre-registered" from the thesis and from your spoken answers. Replace it with an exact statement: which criterion was fixed on which date, and that the final version was applied identically to all five predictors. Also state plainly that the verdict does not change under any of the three versions (it does not: every predictor already fails "fires before grok in 5/5" or the tracking test).

"The criteria were refined during the pilot phase on L2 Norm. The final two tests were fixed before the 5-seed Nanda-Unified run and applied identically to all five predictors. Under every earlier version of the criteria, the verdicts are the same."

2.3 "Your scope says 'published method only', but you imposed your own detection rule."

High

Partly defended

Attack. Nanda et al. describe the weight norm only descriptively. Papyan et al. describe NC1 as a terminal-phase phenomenon. Martin and Mahoney use α to compare trained models, not to time an event. Canatar et al. give a theory of learning curves. None of them defines "the epoch of the extreme value" as a predictor of grok. So the "extreme value" rule is your design. Then how is this a benchmark of *published* predictors?

Where we stand. The 25 September scope decision forbids new detection rules. But a timing test needs some event. The extreme-value rule is the least-assumption choice (no threshold, no window), and Salah and Yevick themselves read the variance *peak*. For the others, the rule is ours. L2 Norm is the extreme case: its MA-crossover and MA-of-MA rules are entirely our own design (windows 50/200/20, skip 100, cutoff 90).

What to do.

- In the thesis, separate two things clearly: the *signal* (taken from the source paper) and the *event rule* (a benchmark-level convention, the same for all predictors, chosen because it has no free parameter).
- Say openly that L2 Norm's rules are pilot-phase rules and are not from a paper. Also report L2 Norm under the same extreme-value convention as the others, so that all five are judged by one rule.
- Optional and threshold-free: the original Gantt plan promised a "Leaderboard: AUC + Lead Time". An AUC-style check (at each checkpoint, can the signal value separate pre-grok from post-grok checkpoints across seeds?) uses no detection rule at all. If the examiner asks "where is the AUC you planned?", you should have an answer ready — either the number, or a clear reason it was dropped.

"Signals are taken exactly from the source papers. Because none of those papers defines a timing event, the benchmark uses one parameter-free convention for all predictors: the epoch of the signal's extreme value. This convention is part of the benchmark, not part of any predictor, and it is applied identically to all five."

2.4 "Your explanations of why each predictor failed — are they tested or just stories?"

Medium

Partly defended

Attack. The defence note gives a mechanism for every failure: Spectral fails because the task is in the feature-learning regime; AGE fails because NC1 is terminal-phase; Dropout fails because robustness comes after circuit formation. Which of these did you measure?

Where we stand. None of the mechanisms has been measured directly. One claim is also inaccurate as written: the 17 September note says "my Phase 1 replication already established grokking on $(a+b) \bmod 113$ is feature-learning". Phase 1 (the M1 gate) was a $p = 97$ single-head reproduction, and it did not measure the lazy/rich regime at all. Worse, the Spectral explanation "no global representation-kernel rank collapse ahead of the jump" is contradicted by the saved eigenvalues (Section 9.2).

What to do. Label every mechanism as "consistent with" rather than "because of". For the regime claim, one cheap measurement from checkpoints will do: the relative weight change $\|W_t - W_0\|_F / \|W_0\|_F$ and the change of the representation kernel over training. Large change supports feature learning. The participation-ratio drop in Section 9.2 is already direct evidence for feature learning, so the regime claim can become stronger, not weaker, once it is measured.

2.5 "Your own documents disagree with each other." Medium Open

Attack. An examiner who opens the repository will look for inconsistencies. Each one costs credibility.

Where we stand. The following inconsistencies exist today:

- The 25 September entries call the v4 benchmark "p = 97, 24-checkpoint". The 7 September v4 entry clearly says the task was $(a+b) \bmod 113$. v4 was p = 113 with 24 checkpoints.
- `context.md` said many times that Nanda et al. use LayerNorm; this was corrected on 3 September (they do not). The Shadow-LayerNorm experiment's motivation rests on the wrong reading. The correction is on record, but the old text is still there (by the append-only rule).
- The yaml says `per_predictor_min_runs: 3` (justified via Power et al.), while the benchmark uses 5 seeds (justified via Nanda et al.).
- The runner's docstring still says "only L2-Norm and Dropout"; `08_Experiments/results/README.md` still describes the old run; the root README objective lists only four predictors.
- The v4 numbers (grok mean 13218.8) and the current numbers (grok mean 12836.8) are both quoted in different places. They are different retrains of the same protocol.

What to do. Append one "errata and canonical numbers" section to `context.md`. In the thesis, quote only the current 5-seed run and mention v4 once, as an earlier 24-checkpoint run with the same verdicts.

3. Angles on the statistical design

3.1 "Five seeds cannot support a correlation claim." Medium Partly defended

Attack. With $n = 5$, a Spearman ρ is significant only at $\rho \geq 0.9$. HTSR's $\rho = 0.90$ is borderline. Any "tracks grok" or "does not track grok" statement is weak.

Where we stand. This limitation is already recorded on 26 September. The verdicts mainly rest on Test 1 (fires before grok in 0/5 seeds for Predictors 2 to 5), and a 0/5 result is strong even with five seeds: if the event were equally likely to fall before or after grok, the chance of 0/5 would be $0.5^5 \approx 0.031$.

What to do. Lead with Test 1 in every answer, and treat ρ as secondary. Report exact permutation p-values and leave-one-seed-out results for every ρ . If compute allows, add seeds (each seed costs about 2.1 hours of training; checkpoint predictors then run without retraining).

"The main test is directional, and every post-hoc predictor fires after grok in five out of five seeds. Under a coin-flip null, that has probability about three percent. The rank correlation is secondary and I report it with its limitation."

3.2 "Your seed changes two things at once." Medium Open

Attack. `torch.manual_seed(seed)` is set, then `random_split` draws the train/test split, then the model is initialised. So each seed has a different data split *and* a different initialisation. The 3.4-fold spread in grok epoch (6921 to 23747) could come from either.

Where we stand. Not separated. The seed-4 outlier (grok 23747, about twice the others) has been "open, to be explained" since 4 September and is still unexplained. It is the extreme point in almost every correlation, and on 4 September one seed alone flipped a correlation from $r = +0.96$ to $r = -0.55$.

What to do. State the confound in the Methods. If you do extra runs, use a factorial design: fixed split with varying init, and fixed init with varying split (2 x 2 or 3 x 3 is enough to show which source dominates). For seed 4, a descriptive look at the saved curves is enough for the colloquium; do not claim a cause.

3.3 "Is MPS training even reproducible?" Low Open

Attack. Grokking is sensitive to small numerical differences. MPS kernels are not guaranteed to be bit-deterministic. If I re-run seed 1, do I get grok at 6921 again?

What to do. Re-run one seed once and report the grok epoch difference. The two retrains of the same protocol already give a partial answer (v4 grok epochs 14474, 6988, 10418, 10193, 24021 vs current 12987, 6921, 9510, 11019, 23747): same seed ordering and similar values, but not identical. That difference itself should be explained in one sentence (the code changed between v4 and now, for example the checkpoint schedule and the extra evaluation work in the loop, so the RNG stream is not guaranteed identical).

4. Angles on the grok definition, timing and resolution

4.1 "Why is grok defined at test accuracy 0.9?" Low Defended

Where we stand. Grok epochs are reported at 0.9, 0.95 and 0.99. There is also a strong one-directional argument: the thresholds 0.9 to 0.99 all sit at the *end* of the transition, so they are the most favourable choice for late signals. If grok were defined at the onset (test accuracy 0.1), the grok epochs would move earlier (seed 0: 11023 instead of 12987; seed 4: 21723 instead of 23747), and every post-hoc predictor would look even later.

"I deliberately used the latest reasonable definitions of grok. Any earlier definition only makes the predictors' lateness worse, so the verdict is robust to this choice."

4.2 "Your events depend on when you stopped training." Medium Open

Attack. NC1 keeps falling, α keeps drifting, alignment keeps rising until the last checkpoint. The "minimum" is then partly "the end of training". Train for 80,000 epochs and the event moves again.

Where we stand. Section 9.3 shows this directly from saved data. For seed 0, the NC1-minimum epoch is 19500 if the run is cut at 20,000 epochs, 29100 at 30,000 and 31500 at 40,000. For seed 1, the NC1 minimum moves from 18700 to 37800. The earlier v4 run already had 3 of 5 NC1 minima pinned at the final checkpoint.

What to do. Report the event epochs at two or three horizons. Point out that the Test 1 verdict does not change, because even with a shorter horizon the events stay after grok whenever grok is inside the horizon. This turns a weakness into a robustness result.

4.3 "The post-grok collapses contaminate your extreme values." Medium Partly defended

Attack. After grok, test *and* train accuracy briefly collapse (seed 1 down to 0.475; 28, 75, 31, 18 and 4 dip epochs in seeds 0 to 4). The dropout-variance peak and the NC1 minimum sit inside this noisy phase. So you may be measuring an optimiser instability, not the predictor.

Where we stand. The explanation (slingshot-type effect of Adam with strong weight decay, Thilak et al. 2022) is recorded as "consistent with, not tested". The weight-norm picture holds for seed 1 but not for seed 4.

What to do. For the colloquium it is enough to show that the events do not coincide epoch-by-epoch with the dip epochs, or, if they do, to say so honestly. The cheap one-seed test (lower learning rate or weight decay, do the dips vanish?) was offered on 25 September and not done; it needs retraining and is optional.

4.4 "A 100-epoch grid cannot resolve lead times." Low Partly defended

Where we stand. The grid choice is derived (at least 6 samples inside the narrowest transition, at most 4% ratio error; save every 50 = gcd of 100, 200, 250). Only predictor events with leads under 100 epochs are affected, and those appear only in the retrospective Result B, not in the verdict.

Open item. The planned sensitivity check (24-point legacy grid vs 100 grid vs 250 grid) is not recorded as done. The transition widths were also supposed to be re-measured on the new run. The onset-to-grok widths on the new run are 1964, 1529, 1585, 2431 and 2024 epochs (from test accuracy 0.1 to 0.9); the narrowest is 1529, very close to the design value of 1570, so the design still holds. Record this.

5. Angles on the substrate and generalisation

5.1 "One task, one prime, one split fraction. Why should anyone believe this generalises?"

Medium

Partly defended

Attack. The whole benchmark uses modular addition, $p = 113$, 30% train, one architecture, one optimiser setting. Power et al. show that grok time depends strongly on the train fraction and on the operation. Salah and Yevick used an MLP on $p = 53$. A predictor could fail here and work elsewhere.

Where we stand. The choice of one faithful Nanda substrate is well argued (fewest moving parts; the substrate the literature standardised on). The 2 September entry already noted that varying p , train fraction and seeds is a better robustness axis than head count; not started.

What to do. For the colloquium, framing is enough: this is a controlled benchmark on the canonical substrate, and its claims are limited to it. If you want one robustness row, the cheapest informative change is the train fraction (for example 0.4), because it moves grok time a lot while keeping everything else fixed. All five predictors then run from checkpoints.

"The claim is deliberately narrow: on the canonical Nanda substrate, under one frozen protocol, none of these predictors leads grokking. I do not claim they fail everywhere."

5.2 "You changed the substrate after the first predictor failed."

Medium

Partly defended

Attack. The project moved from single-head $p = 97$ to four-head $p = 97$, then to $p = 113$ with a new initialisation and new Adam betas — all after L2 Norm had failed. This looks like searching for a setup where things work.

Where we stand. Each change was motivated by fidelity to Nanda et al., and each is documented with its reason (the $N(0, 1)$ embedding transient, the $\beta_2 = 0.999$ slingshot spike). Most importantly, the change did not rescue any predictor: L2 Norm failed on all three substrates. So the history shows the opposite of cherry-picking.

"Every substrate change moved the setup closer to Nanda et al., and each was made for a documented fidelity reason. None of them changed a verdict; L2 Norm failed on all three substrates."

5.3 "Are the remaining deviations from Nanda harmless?"

Low

Defended

Remaining known deviations: output vocabulary 114 (the "=" token is one unused output class), MPS instead of CUDA, and 40,000 epochs over a random 30% split exactly as in Nanda. The weight-norm curve reproduces Nanda's Figure 7 shape across seeds, and grok timing sits in Nanda's range for four of five seeds. Keep this table ready.

6. Angles on each predictor's implementation

6.1 L2 Norm (Predictor 1)

Attack A — "Whose predictor is this?" Nanda et al. do not propose the weight norm as a predictor and say timing cannot be predicted ex ante. The weight-norm-as-controlling-quantity line is Liu et al. (Omnigrok, ICLR 2023, and "Towards Understanding Grokking", 2022). Neither paper is in `04_Literature/Papers/`, although adding them was a recorded to-do since 3 September.

Attack B — "Your rules are home-made." MA-crossover and MA-of-MA zero-crossing were designed and tuned by us in the pilot phase (five strategies). This is exactly the kind of predictor-specific tuning the later scope rule forbids.

Attack C — "Which weights?" The norm includes embeddings, unembedding and MLP biases. Per-module $\sum w^2$ is logged for 5 groups but not analysed in the verdict.

What to do. Add Liu et al. to the literature and cite them as the source of the idea. Report L2 under the same extreme-value convention as the rest. Show the per-module curves once, as a descriptive figure. Use Jonathan's own recorded account (first predictor, pilot-phase effort) if asked why L2 has more machinery than the others.

"L2 Norm was my pilot predictor. Its extra rules are pilot-phase engineering, not a published method, and they are reported as such. Under every rule I tried, it fires at an epoch that does not move with grok: 113 to 137, and about 1800, while grok ranges from 6921 to 23747."

6.2 Dropout (Predictor 2)

Attack A — "Your variance peak is just a coin-toss statistic." High At one checkpoint, each pass samples fresh dropout masks. For fixed weights, each test example is correct with some probability q_i , independently of other examples. Then the variance of the pass accuracy is

$$\text{Var}(\text{acc}) = \frac{1}{n^2} \sum_{i=1}^n q_i(1 - q_i) \leq \frac{m(1 - m)}{n},$$

where m is the mean dropout accuracy and $n = 8939$ test examples. This bound is largest at $m = 0.5$, where it equals about 2.8×10^{-5} . The observed peak values are 3.4 to 3.9×10^{-5} , the same order. Section 9.1 shows that the measured variance correlates with $m(1 - m)/n$ at $r = 0.95$ to 0.98 in every seed. In plain words: the "variance signal" is almost fully decided by where the mean dropout accuracy sits. Its peak is roughly "the checkpoint where mean dropout accuracy is near 0.5", plus sampling noise from 100 passes. The observed peaks slightly above the bound are consistent with that sampling noise.

Why this matters. Salah and Yevick interpret the variance as a separate signal. On our substrate, it carries little information beyond the mean dropout accuracy. This is not a bug in our code; it is a property of the statistic. It also explains why the peak arrives late: mean dropout accuracy at rate 0.5 reaches the middle range only after the circuit exists.

What to do. Put the binomial comparison in the thesis as a diagnostic, with one figure per seed (measured variance vs $m(1 - m)/n$). This strengthens the negative verdict and pre-empts the question. Do not call it a new predictor.

"I checked the variance against a binomial null. With independent dropout masks per example, the variance is bounded by $m(1-m)/n$, and my measured variance follows that curve with correlation above 0.95 in every seed. So on this substrate the variance adds little beyond the mean robustness, and its peak comes after grok."

Say this only after the check is re-done inside the project and recorded.

Attack B — "Dropout is in the wrong place." Medium Our dropout acts only on the attention-branch output and the MLP-branch output (two residual branches). The residual stream from the embeddings is never dropped. Salah and Yevick apply dropout inside a two-layer MLP (256 hidden units, $p = 53$). So "robustness to dropout" means a different thing here. Also the rate is 0.5, not the 0.3 of their Figure 1 sweep.

What to do. State placement and rate as documented deviations. A single extra run at rate 0.3 from saved checkpoints (dropout variance is checkpoint-computable) would close the rate question. Note: the checkpoint recompute path currently rebuilds the test loader without re-seeding (Section 8.1); fix that before any such recompute.

Attack C — "The dropout gap is only measured at the final epoch." Low Then it cannot be a timing predictor at all. Answer: correct, and it is reported only as a post-hoc characterisation; the per-checkpoint DRC snapshots exist if needed.

6.3 Spectral — Canatar task-model alignment (Predictor 3)

Attack A — "Your k_{90} lives in the null space." High The kernel is $K = \Phi_c \Phi_c^\top$ with $\Phi \in \mathbb{R}^{3830 \times 128}$. So $\text{rank}(K) \leq 128$ (at most 127 after centring). About 3700 eigenvalues are numerically zero. Inside that zero eigenspace, the eigenvectors returned by `eigh` are an arbitrary basis, and their order is arbitrary too. The cumulative task power $C(k)$ reaches only about 0.14 to 0.22 within the top 100 modes at the end of training (Section 9.2). So 80% or more of the label power sits in the null space, and $k_{90} \approx 3400$ is decided by how `eigh` happens to order the null-space vectors. The statement " k_{90} falls only 2–3%, so no concentration" is therefore not a valid reading of the data.

Attack B — "Actually, the spectrum does collapse around grok." High The saved top-50 eigenvalues tell a different story. The participation ratio $\text{PR} = (\sum_k \eta_k)^2 / \sum_k \eta_k^2$ over the top 50 eigenvalues rises from about 30 at initialisation to about 36–37 at epoch 1000, then falls to about 9–16 at the grok epoch, and to about 3–9 at the end. In all five seeds it is still fairly high (about 19–32) at 80% of the grok epoch. This contradicts the recorded explanation "no global representation-kernel rank collapse ahead of the jump". What the data suggests is: a strong rank collapse, roughly *concurrent* with grok.

Is the verdict in danger? Not necessarily. The verdict rests on the extreme-value events (alignment maximum, k_{90} minimum) landing after grok. The alignment score only uses the non-zero modes, so it is not affected by the null-space issue. But the *explanation* of the failure must be rewritten, and the k_{90} rows should be dropped or recomputed restricted to the non-null subspace. Otherwise the examiner can show one plot and break the story.

Attack C — "Canatar is a regression theory; you train with cross-entropy." Medium Canatar et al. derive generalisation error for kernel *regression* with squared loss, and use the NTK of infinitely wide networks. We use a last-layer representation kernel (a conjugate kernel, not the NTK) of a network trained with cross-entropy, and project centred one-hot labels. A linear regressor on Φ explains only about 14–22% of the one-hot label variance, although argmax accuracy is 100%. That gap is exactly the regression-vs-classification difference.

Attack D — "Kumar et al. connect grokking to kernel alignment directly." Medium Kumar, Bordelon, Gershman and Pehlevan (ICLR 2024; Pehlevan is a co-author of the Canatar paper) describe grokking as a transition from lazy to rich training, with NTK/kernel alignment changing through it. An examiner may ask why the Spectral verdict leans on "Canatar does not apply here" rather than engaging with this line.

What to do. (1) Recompute $C(k)$ and k_{90} on the non-null subspace only (for example normalise by the power captured in the top r modes, $r = \text{rank}(K)$), from checkpoints. (2) Replace the "no rank collapse" explanation with the participation-ratio curve and the honest reading: the collapse is roughly concurrent with grok, so it does not lead. (3) State the conjugate-kernel and cross-entropy deviations. (4) Cite Kumar et al. as related work.

"The representation kernel has rank at most 128, so I read the spectrum only in its non-zero part. There, the representation does compress strongly — the participation ratio falls from about 36 to about 10 — but this compression happens together with the test-accuracy jump, not ahead of it. That is why the Canatar alignment signal does not lead grok."

Say this only after the non-null recomputation is done and recorded.

6.4 AGE — NC1 variability collapse (Predictor 4)

Attack A — "That is not Papyan's NC1." Papyan et al. define $\mathcal{NC}_1 = \frac{1}{C} \text{Tr}(\Sigma_W \Sigma_B^\dagger)$. The project uses the trace ratio $\text{Tr}(\Sigma_W) / \text{Tr}(\Sigma_B)$ with equal class weighting. The trace ratio is a common simplification, but it is not the paper's formula. The two can behave differently when Σ_B is badly conditioned, which is exactly the situation when the representation rank collapses (Section 9.2).

What to do. Compute Papyan's exact form from the same checkpoints (about 20 seconds per seed) and report both. If the verdict is the same, the attack is closed.

Attack B — "Where does the name AGE come from?" "Adaptive Grokking Epoch via Neural Collapse" is the project's own label; the definition was locked on 6 September. Make sure that this matches what the supervisor meant by "AGE" in the original four-predictor baseline. If the name came from the earlier baseline proposal with a different meaning, the examiner may ask.

Attack C — "Why the minimum and not the onset?" Already recorded as not tried (NC1 < 1 onset). Under the scope rule it should stay not tried; say that the extreme-value convention is common to all predictors (Section 2.3). The retrospective Result B already shows that NC1 has moved substantially before grok; present that only as a descriptive observation, as agreed on 25 September.

"NC1 falls about 660-fold in every seed, but, as Papyan et al. describe it, it is a terminal-phase quantity. Its minimum arrives after grok in five of five seeds, with both the trace-ratio and the original pseudo-inverse definition."

The last clause only after the exact-form check.

6.5 HTSR Alpha (Predictor 5)

Attack A — "Your matrices are too small for power-law fits." Each layer has at most 128 eigenvalues (114 for the unembedding). The automatic $x_{\min}$ sometimes keeps a tail of only 4 eigenvalues. Fitted α is below 2 in 85–94% of fits, outside Martin and Mahoney's usual 2–6 range.

Attack B — "Did you test that it is a power law at all?" The Clauset–Shalizi–Newman procedure (which the `powerlaw` package implements) includes likelihood-ratio comparison against alternatives such as exponential, log-normal and truncated power law. Only the fit is done now, not the comparison.

Attack C — "Why exclude the embeddings?" Nanda et al.'s Fourier structure is clearest in the embedding matrix. HTSR analyses only the six `nn.Linear` matrices. The reason (WeightWatcher analyses linear and convolutional layers) is fair, but an examiner may still ask.

What to do. Add `distribution_compare` results per layer from checkpoints (cheap, 0.25 s per checkpoint). Report the fraction of fits where the power law is not preferred. Keep the $\alpha < 2$ caveat, and state that nothing was tuned to move α . An optional one-line sensitivity with embeddings included can close Attack C.

7. Angles on missing comparators

7.1 "Why did you not compare against X?" Medium Open

These are the names an informed examiner is most likely to raise. None of them is a new idea; all are published.

Comparator	Why an examiner may raise it	Suggested handling
Nanda et al. (2023): restricted loss, excluded loss, Gini of Fourier components	The best-known "progress measures" for this exact substrate, from the paper whose substrate you copied	Use as positive control (Section 2.1), or explain clearly why they are not predictors (they need the reverse-engineered circuit)
Notsawo et al. (2023), "Predicting Grokking Long Before it Happens" (ICLR 2024)	Title says it all: a published grokking-specific predictor based on the early loss curve	Mention in related work; note it is not in the nine-predictor list chosen with the supervisor
Kumar, Bordelon, Gershman, Pehlevan (ICLR 2024), "Grokking as the Transition from Lazy to Rich Training Dynamics"	Directly relevant to the Spectral explanation	Cite in the Spectral discussion (Section 6.3)
Liu et al. (2022, 2023), "Towards Understanding Grokking" and "Omnigrok"	The real source of weight norm as the controlling quantity	Add to literature; cite for L2 Norm (Section 6.1)
Trivial baselines: train loss, "time since memorisation", fixed epoch	Every benchmark needs a floor	Memorisation (train accuracy 0.99) happens at epoch 205–213 in all seeds, while grok ranges 6921–23747. So "time since memorisation" is useless — a quick, strong sentence for the thesis

"The nine predictors were fixed with my supervisor at the start. Nanda's progress measures are used as a positive control, not as a benchmark row, because they need the reverse-engineered circuit and are not general predictors."

8. Angles on code and reproducibility

8.1 Latent bug: checkpoint recompute rebuilds the test split without re-seeding Medium

[Open](#)

What is wrong. In `recompute_from_checkpoints()` (`run_nanda_benchmark.py`, around line 1042), the test loader is built with `get_data loaders(...)` *without* `torch.manual_seed(seed)` first. `random_split` then draws a different split from the one the model was trained on. The "test" set would then contain training examples.

Does it affect current results? No, as far as the record shows. Dropout variance was computed live during training in the current run, and Spectral, AGE and HTSR re-seed their own loaders or need no data. It becomes real the moment anyone recomputes dropout variance from checkpoints (for example the rate-0.3 check in Section 6.2).

What to do. Fix it before any dropout recompute, and record it. An examiner who finds it first will ask whether any number used it; you should be able to say "no, and here is why".

8.2 Smaller code and record items Low

- Train accuracy is computed from the logits of the forward pass *before* the optimiser step, and test accuracy after it: a one-step offset. Harmless, but know it.
- `compute_dropout_variance` still has default `n_samples=30`; the runner always passes 100.
- An uncommitted edit to `context.md` was lost on 26 September; its content is unknown. If asked about the append-only record, say this once, honestly.
- The repository carries about 3 GB of checkpoints in git history. Good for transparency; make sure the examiners know where the results are, so they do not clone the whole thing unprepared.
- The v4 plotting script `plot_nanda_results.py` was deleted; the v4 reproduce command therefore needs a restore from the tag.

9. The quick checks run for this report

All three checks read saved arrays under `08_Experiments/results/nanda_unified/seed_*/`. Nothing was written to the project. The checks are descriptive and must be re-done inside the project (as a script, recorded in `context.md`) before being quoted.

9.1 Dropout variance vs the binomial bound

Quantity compared: measured variance of accuracy over 100 dropout passes (rate 0.5) vs $m(1 - m)/n$, with m the mean dropout accuracy at the same checkpoint and $n = 8939$.

Seed	Grok	Variance peak epoch	Peak value	Mean dropout acc at peak	Epoch where $m(1 - m)$ is largest	corr(variance, $m(1 - m)/n$)	Median ratio variance / bound
0	12987	24200	3.69e-5	0.412	16100	0.968	0.95
1	6921	25400	3.36e-5	0.321	9200	0.952	0.95
2	9510	16700	3.87e-5	0.705	38200	0.955	0.96
3	11019	18800	3.88e-5	0.563	24800	0.965	0.98
4	23747	39000	3.56e-5	0.479	28000	0.979	0.99

Reading. The variance follows the binomial curve closely (correlation 0.95–0.98, variance about 95–99% of the bound in the median). The exact peak epoch sits on a broad, flat maximum, so it is decided by sampling noise; that is why it does not coincide exactly with the epoch where $m(1 - m)$ peaks. Either way, both lie after grok in every seed.

9.2 Spectral: null space and participation ratio

(a) Cumulative task power in the top 100 kernel modes, $C(100)$, at the first, middle and last evaluated checkpoint:

Seed	First	Middle ($\approx$ epoch 20000)	Last (epoch 39999)	Final k_{90} (recorded)
0	0.020	0.174	0.171	3368
1	0.020	0.146	0.140	3390
2	0.020	0.151	0.136	3406
3	0.020	0.195	0.219	3336
4	0.020	0.113	0.183	3364

Since $\text{rank}(K) \leq 128$, the power beyond mode 128 lies in the zero eigenspace, whose ordering is arbitrary. So k_{90} in the thousands is not interpretable.

(b) Participation ratio of the top 50 kernel eigenvalues, $\text{PR} = (\sum \eta_k)^2 / \sum \eta_k^2$, at selected epochs (nearest evaluated checkpoint):

Seed	Grok	Epoch 0	Epoch 1000	$0.5 \times \text{grok}$	$0.8 \times \text{grok}$	At grok	$1.2 \times \text{grok}$	Epoch 39999
0	12987	29.8	37.0	23.0	19.2	9.2	5.8	4.4
1	6921	30.1	36.2	9.1	25.2	9.8	4.9	3.3
2	9510	30.2	37.2	35.6	31.7	15.7	8.9	7.0
3	11019	30.4	36.6	23.9	20.5	11.6	10.1	8.5
4	23747	30.5	36.0	33.0	31.4	8.7	5.3	5.8

Reading. The representation compresses strongly, and a large part of that compression happens close to grok. Seed 1 is not monotone (9.1 at half its grok epoch, then 25.2), so the curve is noisy between checkpoints. This is a descriptive observation over only the saved top-50 eigenvalues, not a predictor, and no rule is proposed from it. Its role in the thesis is to replace an incorrect explanation, not to rescue Spectral.

9.3 Horizon dependence of the extreme-value events

Event epoch if the run had stopped at 20,000 / 30,000 / 40,000 epochs (same saved checkpoints, restricted to epochs below the horizon):

Seed	Grok	NC1 minimum	$\bar{\alpha}$ minimum	Alignment maximum	Dropout-variance peak
0	12987	19500 / 29100 / 31500	19100 / 19100 / 34300	19400 / 29100 / 31600	13700 / 24200 / 24200
1	6921	18700 / 18700 / 37800	16000 / 16000 / 16000	18700 / 18700 / 37800	16600 / 25400 / 25400
2	9510	15400 / 29300 / 29300	17900 / 24100 / 24100	15400 / 29300 / 29300	16700 / 16700 / 16700
3	11019	18400 / 26800 / 37400	19500 / 19500 / 19500	18400 / 26800 / 37400	18800 / 18800 / 18800
4	23747	19600 / 29900 / 30100	14400 / 28500 / 39600	19600 / 29900 / 32000	19700 / 27900 / 39000

Reading. The events move with the horizon, often by 10,000 epochs or more. Whenever grok lies inside the horizon, every event still lies after grok. (For seed 4 at the 20,000 horizon, grok itself is outside the window, so that column is not a test.) Two further points worth noting: the NC1 minimum and the alignment maximum coincide at the same checkpoint in many cells; this was already seen on 26 September for three seeds and is still unexplained.

10. Suggested order of work before the colloquium

Everything in the first block needs no retraining. It uses the 801 saved checkpoints per seed or the saved arrays.

Step	Action	Closes	Needs
1	Wording fixes: remove "pre-registered"; separate signal vs event rule; "consistent with" for every mechanism; errata section in <code>context.md</code>	2.2, 2.3, 2.4, 2.5	Writing only
2	Positive and trivial controls: test loss (and optionally Nanda's excluded loss) and a fixed-epoch rule through the same two tests	2.1, 7.1	Checkpoints
3	Dropout binomial diagnostic as a proper script and figure	6.2 A	Saved arrays
4	Spectral: non-null $C(k)$, participation-ratio figure, rewrite the explanation	6.3 A, B	Checkpoints
5	AGE with Papyan's exact $\mathcal{NC}_1$; HTSR <code>distribution_compare</code>	6.4 A, 6.5 B	Checkpoints
6	Horizon table; grid sensitivity (24 vs 100 vs 250); re-measured transition widths recorded	4.2, 4.4	Saved arrays / checkpoints
7	Fix the recompute test-split seeding bug; then optional dropout rate 0.3	8.1, 6.2 B	Code + checkpoints
8	Literature: add Liu et al. (both), Kumar et al., Notsawo et al.	6.1 A, 6.3 D, 7.1	Reading

The second block needs retraining and is optional, depending on time and the supervisor's advice: one MPS reproducibility re-run of a single seed (3.3); a small split-vs-init factorial (3.2); one robustness row at a different train fraction (5.1); one low-learning-rate run for the post-grok dips (4.3).

A last point. Per the project rule, Predictor 6 (Correlation Traps) is next in the evaluation order. The steps above are checks of finished work, and most of them are cheap. It is worth discussing with Prof. Rashid which of them he wants before moving on, because steps 2 to 4 change how the first five results are explained in the thesis.

References mentioned

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